



MEDILINK OS

Healthcare Outpatient Operations & Clinical Intelligence Platform

Executive Summary: MediLink OS is a next-generation clinical coordination and outpatient nervous system designed to orchestrate hospital environments. It bridges patient journeys, medical documentation, real-time telemetry, and billing administration into a unified, secure ecosystem. Optimized for mobile-responsive environments and high-availability operations, it enforces HIPAA compliance, remedies queue bottlenecks, mitigates drug interaction hazards, and automates patient bed turnaround times.

Document ID:	ML-OS-ARCH-2026	Tech Stack:	Next.js 16 / React 19 / Postgres
Version:	v2.4.0 (Latest Release)	Database ORM:	Prisma Schema v7.8
Deployment:	Vercel Enterprise	Real-Time Engine:	Pusher Websockets
Publish Date:	June 20, 2026	Security:	HIPAA Compliant Audit Trail

System Architecture & Tech Stack

MediLink OS is built upon a modular, highly performant SaaS stack structured to withstand high-concurrency clinic operations and real-time telemetry updates. The application utilizes a hybrid SSR (Server-Side Rendering) and CSR (Client-Side Rendering) framework to deliver instantaneous page updates.

Architecture Layer	Technology Selection	Operational Impact
Frontend Framework	Next.js 16.2 / React 19.2	Leverages Server Components for speed & client-side portals for mobile takeovers.
State Management	Zustand 5.0	Ultra-lightweight state store syncing cache without React context overhead.
Database & Engine	PostgreSQL (Neon Serverless)	Serverless database clustering with connection pooling, autoscaling, and Neon branch environments.
Database ORM	Prisma ORM 7.8	Type-safe client generation, complex relations, and automated migration scripts.
Real-Time Sync	Pusher WebSockets Engine	Bi-directional instant event broadcast (reminders, queue shifts, and consent changes).
Identity & Auth	NextAuth v5 (Beta 31) / Credentials	Role-Based Access Control (RBAC) supporting PATIENT, DOCTOR, ADMIN, NURSE, etc.
Mobile Integration	Capacitor Core 6.0	Wraps the Next.js web application into native iOS and Android binaries with haptic feedback.
AI Scribe & Auditing	Groq SDK (Llama-3 models)	Instantaneous JSON structuring of ambient voice transcripts into structured SOAP notes.

Core Architecture Hardening & Latest Backend Updates

- **RBAC Middleware Shielding:** The middleware layer intercepts all API requests (``/api/beds``, ``/api/consent``, ``/api/audit``, etc.) to enforce strict JWT authentication and role validation, preventing unauthorized vertical privilege escalation.
- **Real-Time Broadcast Webhooks:** Integrated Twilio WhatsApp incoming webhook handler (``/api/messages/trigger``) linked with Pusher. This lets patients log medicine compliance by hitting single-character replies in WhatsApp, triggering real-time UI updates on the doctor and patient dashboards.
- **DB Performance Indexing:** Optimized PostgreSQL table lookups with index additions on ``Visit(status)``, ``Visit(doctorId, status)``, ``AuditLog(patientId, timestamp)``, ``PatientConsent(patientId, status)``, and ``SnoozedReminder(status, scheduledAt)``.

Hospital Mission Control (Admin Portal)

The Administrative console acts as the command center for clinic operations. It is designed to provide real-time situational awareness, automate patient transitions, and secure HIPAA compliance.

Key Capabilities & Workflow Systems:

1. Interactive Floor Map & SLA Bottleneck Alerts:

Integrates a live vector blueprint of the clinic (Triage, Doctor Rooms, Pharmacy, Cashier Billing). The dashboard runs real-time queue delay calculations (SLA metrics). If Dr. Patel's wait queue exceeds 15 minutes, Room 403 flashes an operational bottleneck warning. Administrators can click any room on the interactive blueprint to dispatch floor manager assistance or trigger reassurance WhatsApp templates.

2. Interactive Ward Bed Occupancy Matrix:

Lists beds in ICU, ER, General, and Isolation wards. Displays bed status (AVAILABLE, OCCUPIED, MAINTENANCE, RESERVED) and equipment indicators (ventilator attachment). Administrators can allocate patient IDs directly to beds, or utilize *Auto-Dispatch*, which automatically processes a patient's live vitals through a Clinical Acuity Triage score (analyzing Systolic BP, HR, SpO2, and Chief Complaint) and dispatches them to the most appropriate ward.

3. Coordinated Patient Discharge & Sanitization Turnaround:

Tracks patient discharge schedules. Admin coordinates with staff to: assign transporter escorts, claim bed sanitization cleaning tasks by crew names, and log sanitization completions. The system calculates the total turnaround time (in minutes) and updates the Ward Bed Matrix instantly to maximize clinic throughput.

4. AI Billing Claims Audit & TPA Submission:

Pre-scans bills using AI prior to insurance submission. The module assesses missing clinical notes, evaluates billing-diagnosis ICD-10 consistency, flags missing documentation, and assigns an AI-driven Rejection Risk score (0-100%) to mitigate claim denials from Third Party Administrators (TPA).

5. Immutable Audit Trail (HIPAA Gateway):

Ensures absolute data compliance. Every administrative action (bed release, clinical note access, consent revocation, or emergency override) is logged onto an immutable audit trail table storing the timestamp, user, role, patient, and detail payload for official security audits.

Doctor Workspace & AI Scribe Engine

The Doctor Workspace optimizes clinical workflows to reduce cognitive fatigue, enforcing safety guidelines while automating tedious EHR documentation tasks.

Key Capabilities & Workflow Systems:

1. Ambient Scribe & Llama-3 AI SOAP Engine:

Doctors activate a hands-free microphone during patient consultations. The system listens in real-time. On completion, it sends the transcript to the AI Scribe API. The engine extracts symptoms, vitals, and treatment plans, structuring it into a compliant SOAP clinical note (Subjective, Objective, Assessment, Plan) and generating structured prescription lists.

2. Hands-Free Voice Command Orchestration:

Equipped with an advanced clinical command interpreter. While recording, the doctor can prefix directions with wake-words, e.g., *'MediLink prescribe Amlodipine 5mg QD for 14 days'*, *'MediLink diagnose Essential Hypertension'*, *'MediLink order lab Holter ECG'*, or *'MediLink sign note and next patient'*. The engine parses these commands on-the-fly, creating prescriptions, mapping diagnoses, and executing sign-offs.

3. Drug-Drug Interaction (DDI) Safety Shield:

During prescription creation, the DDI Safety Shield references the patient's active chronic medications (from their ABHA Health Locker) and cross-audits the new prescriptions for contraindications. It checks for critical interactions like:

- *Antihypertensive Antagonism*: Amlodipine + acute Ibuprofen (reduces BP efficacy).
- *Serotonin Syndrome*: Sumatriptan + chronic Sertraline (severe hazard).
- *Duplicative NSAID Therapy*: Simultaneous NSAIDs (renal and GI ulcer hazard).

The system alerts the doctor with an automated safety audit score (0-100) and offers one-click 'Auto-Swap' resolutions (e.g., swapping Ibuprofen with safe Paracetamol, or Sumatriptan with NSAID monotherapy).

4. Cryptographic ABHA Consent & Break-Glass Gateway:

To view notes or test results, doctors must verify active patient authorization. If absent, they can request consent directly (notifying the patient on WhatsApp/EHR). In critical conditions, the doctor can activate an emergency 'Break Glass' bypass, granting immediate access. The system rings a clinical alert chime, triggers native haptics, and logs an immutable entry to the HIPAA audit ledger containing the doctor's credentials.

5. Doctor Wellness Alerts:

Monitors consultation counts and times. An integrated sensor alerts the doctor to suggest short, structured breaks after successive consulting sessions to mitigate professional burnout.

Patient Portal & Aetheris Longevity Autopilot

The Patient Portal coordinates the patient's physical clinic visit while offering tools for chronic care, wellness tracking, and automated medication adherence.

Key Capabilities & Workflow Systems:

1. Interactive Journey Queue & Wayfinding:

Displays real-time queue standings, wait metrics, and delays. When a user changes locations, double-tone wayfinding chimes ring to guide the patient from Entrance to Triage, Doctor Room 402, and Pharmacy.

2. Coordinated Discharge Gateway HUD:

Patients occupying a ward bed can view their discharge progress directly. The HUD details billing approval status, doctor sign-offs, and transporter assignment details, eliminating wait times during discharge.

3. Simplify View (Stress Reduction Mode):

To ease patient anxiety, users can toggle 'Simplify View'. This enlarges elements, increases spacing, and hides clinical graphs, replacing them with reassuring progress messages.

4. Pathology Explainer (AI Lab Translation):

Translates medical lab values into plain English. It highlights parameters, alerts severity levels, and suggests lifestyle next steps.

5. Pharmacy Adherence & Snooze Systems:

Pusher synchronization links patient medicine lists with twilio/whatsapp scheduling. Patients receive WhatsApp reminders. They can snooze reminders for 15 minutes or mark them as taken, which syncs back to the EHR.

6. MediBuddy Chatbot & Dietitian Portal:

Patients can consult MediBuddy for questions, DASH diet advice, and side effect support. It also facilitates booking clinical dietitians.

7. Aetheris Longevity Autopilot:

A premium longevity suite designed for desktop and mobile 393x852 viewports. Features include:

- *Biological Twin*: Computes biological age based on stress, sleep, and nutrition.
- *Exposome Shield*: Integrates climate indices to adjust supplement compounding formulas.
- *Biometric Synthesizer*: Tracks telemetries and flags anomalies.
- *Smart Supplement Cabinet*: Displays custom compounding supplement formulas.
- *Voice Fatigue & Squat Scanner*: Runs AI scans via camera/mic, giving posture tips and longevity points.

Database Architecture & Schema Relations

The MediLink OS database utilizes a PostgreSQL database managed via Prisma ORM. The schema is designed to enforce relational integrity, data privacy (HIPAA compliance), and low latency through explicit indices on key search vectors.

Database Model Schemas & Key Fields:

Model	Key Fields	Relationships & Constraints
User	id, email, password, role	Optional links to Patient (patientId) or Doctor (doctorId) for profile details.
Patient	id, name, phone, abhald, bloodGroup	Linked to User, PatientConsent records, WardBed, and active Visits.
Doctor	id, name, specialization, department	Linked to User and Doctor-Patient Visits.
Visit	id, patientId, doctorId, status, queuePosition, vitals	Core transaction model. Cascade link to ClinicalNote and BillingClaim.
ClinicalNote	id, visitId, subjective, assessment, medications	Unique link to Visit. Medications stored as Json (array of structures).
PatientConsent	id, patientId, accessorId, accessLevel, status, expiresAt	Stores verifiable signed HMAC tokens for clinical access control.
AuditLog	id, userId, action, patientId, timestamp	Immutable HIPAA audit ledger for clinical data access and overrides.
BillingClaim	id, visitId, amount, status, rejectionRisk	Unique link to Visit. Stores AI-flagged missing docs and rejection probability.

Core Relational Flows:

- Patient Registration & Queue:** Creating a Visit links Patient and Doctor. Atomic increment of queuePosition occurs on check-in.
- Clinical Consultation Scribe:** Voice script calls generate a ClinicalNote associated with a unique Visit. Prescriptions sync dynamically to the medications field.
- HIPAA Audit & Break-Glass:** Prior to data fetch, PatientConsent is checked. If missing, Emergency Override bypasses it, writing an AuditLog.
- Intelligent Claim Audit:** Completing a consultation triggers a BillingClaim. Groq audits SOAP logs and flags rejection risks.